		score	%gram.	#dif.wds.	#unq.wds.	%l.d.t.	#unq.n.cks.	#unq.q.m.a.	fl.-kinc.	w2v	av.s.ln.	holistic
Lng.	gen. lin.	0.695	0.584	0.514	0.400	0.493	0.527	0.259	0.258	0.143	0.765
	gramm.	**0.698**	0.505	0.469	0.370	0.423	0.468	0.189	0.265	0.134	0.737
	orth.	**0.718**	0.395	0.317	0.244	0.291	0.350	0.155	0.206	0.073	0.652
	voc. ctrl.	0.652	**0.638**	**0.580**	**0.445**	0.537	**0.600**	**0.263**	0.291	**0.189**	0.779
	voc. rg.	0.651	**0.621**	0.568	0.424	**0.548**	0.576	0.254	**0.339**	0.177	0.749
Prg.	propos.	0.601	0.607	0.545	0.389	0.528	0.568	**0.294**	**0.351**	**0.202**	0.702
	coh.	0.662	**0.621**	**0.574**	0.410	**0.551**	**0.588**	0.248	0.336	0.180	0.774
	flexib.	0.424	0.414	0.390	0.291	0.367	0.412	0.178	0.195	0.125	0.443
	themat.	0.584	0.544	0.527	**0.428**	0.516	0.534	0.203	0.287	0.145	0.650
	holistic	0.732	0.640	0.665	0.451	0.623	0.637	0.178	0.364	0.141	1.000

Table 3: SRC correlation of the GPT-4 predicted scores and relevant linguistic features (**using holistic scores predicted by the Longformer-based grader**). The *holistic* entry refers to the ground-truth holistic scores. In bold the two highest correlations columnwise.

		score	%gram.	#dif.wds.	#unq.wds.	%l.d.t.	#unq.n.cks.	#unq.q.m.a.	fl.-kinc.	w2v	av.s.ln.	holistic
Lng.	gen. lin.	0.726	0.574	0.541	0.414	0.522	0.519	0.197	0.267	0.129	0.814
	gramm.	**0.731**	0.472	0.464	0.363	0.433	0.450	0.100	0.286	0.030	0.791
	orth.	**0.726**	0.436	0.398	0.310	0.354	0.427	0.146	0.203	0.060	0.729
	voc. ctrl.	0.674	**0.640**	**0.621**	**0.453**	**0.591**	**0.624**	**0.243**	0.319	0.179	0.854
	voc. rg.	0.672	**0.624**	**0.582**	**0.452**	**0.563**	0.573	0.218	0.280	0.134	0.816
Prg.	propos.	0.600	**0.624**	0.581	0.417	0.560	**0.593**	**0.261**	**0.353**	**0.190**	0.771
	coh.	0.702	0.555	0.534	0.372	0.511	0.535	0.238	**0.339**	**0.201**	0.827
	flexib.	0.425	0.370	0.368	0.249	0.357	0.368	0.140	0.163	0.104	0.488
	themat.	0.639	0.514	0.504	0.413	0.492	0.483	0.224	0.264	0.179	0.745
	holistic	0.732	0.640	0.665	0.451	0.623	0.637	0.178	0.364	0.141	1.000

Table 4: SRC correlation of the GPT-4 predicted scores and relevant linguistic features (**using ground truth holistic scores**). The *holistic* entry refers to the ground-truth holistic scores. In bold the two highest correlations columnwise.

linguistic range score, which is a broad indicator by definition since it includes elements of grammatical accuracy, syntactic complexity, and vocabulary, and, as a result, shows strong correlations with multiple features. On the other hand, the aspect of flexibility seems to be a little problematic with respect to both the features and the holistic score, probably also due to its "longitudinality", since it seems to be evaluated in relation to previous performances, according to its descriptors (see Appendix A).

Finally, we selected some essays in which there was a large discrepancy between two or more analytic scores, and we evaluated them impressionistically. One example can be found in Appendix B. If we focus on the highest and lowest scores, we notice vocabulary range and orthographic control on one hand, and coherence and cohesion on the other hand. Although quite extreme, this discrepancy makes sense, considering that the learner uses almost no connectors at all and mostly uses coordinating clauses (or even parataxis), but has quite a rich vocabulary and makes no orthographic errors (except for punctuation).

5.4 Statistical tests

Additionally, we explore the relationships among analytic scores using a repeated measures design in order to assess whether there are significant differences among them. While the repeated measures analysis of variance (rANOVA) is a widely known approach for such designs, our data fail to meet the assumptions of sphericity and normality required for its application. Hence, we employ the Friedman test (Friedman, 1937), known as the non-parametric equivalent of rANOVA. This test assesses whether there are significant differences in ranks among multiple paired groups. With a significant p-value obtained, we confirm significant differences among the analytic scores. To determine which scores show significant differences, we conduct post-hoc multiple comparisons using the Ne-

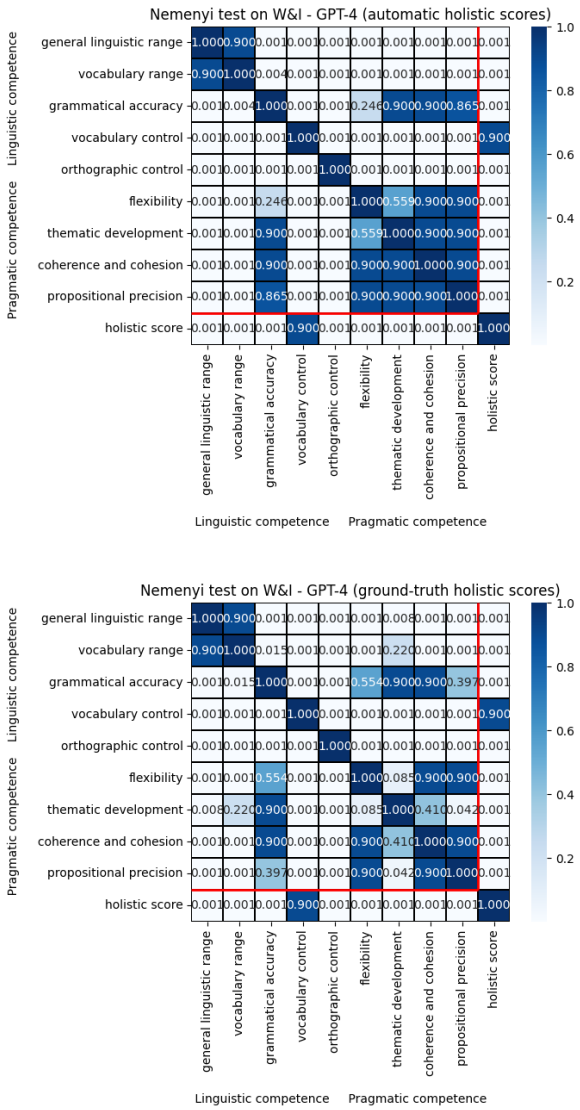

Figure 2: Results of the post-hoc Nemenyi test.

menyi test (Nemenyi, 1963), whose results are reported in Figure 2. The majority of the paired comparisons, even those with the holistic score (except when paired with vocabulary control), show significant differences (i.e., p-value<0.05) both when we provide the ground truth and the automatic holistic scores to GPT-4. In addition to the pairs "general linguistic range - vocabulary range" and "thematic development - vocabulary range", which have some clear overlaps in their descriptors, there seem be non-significant differences over the group of aspects related to the pragmatic competence (i.e., flexibility, thematic development, coherence and cohesion, and propositional precision) and the aspect of grammatical accuracy. While we could expect to see non-significant differences among the aspects related to the pragmatic competence due to their frequent overlaps, the non-significant differences of these with grammatical accuracy might be explained with the fact that not only do its descriptors stress the importance of correctness but, as shown in Appendix A, they also emphasise complexity (e.g., for A1: "Shows only limited control of a few simple grammatical structures [...]"; for B2: "Has a good command of simple language structures and some complex grammatical forms [...]"), which is inherently connected to aspects such as thematic development and coherence and cohesion (Purpura, 2004). In this regard, it is also worth noting that the coherence and cohesion score is the third most correlated with grammatical error rate.

To sum up, under ideal conditions, GPT-4 appears to produce analytic scores that are very reasonably related to the proficiency aspects they are expected to evaluate. The fully-automated pipeline is not always consistent with the ideal system but generates results that are mostly in line with it. This is especially evident for the scores pertaining to grammar and vocabulary.

6 Conclusions

In this paper, we have conducted an initial study on the use of GPT-4 for assessing 9 individual aspects of L2 writing underlying the CEFR proficiency levels in a zero-shot fashion. To do this, we used a holistic grading system on the essays of the W&I validation set and, subsequently, fed them with their respective holistic scores into GPT-4, asking to assess one individual aspect at a time. Although the ground truth analytic scores are not available, we have obtained significant correlations between the predicted analytic scores and various features linked to the componential aspects of the CEFR levels. Beyond its immediate implications for computer-assisted language learning applications, we believe that our exploratory experiments may hold promise as valuable contributions to theoretical studies on construct validity in the broader field of language testing and assessment, given the inclusion of CEFR descriptors in our study.

In order to collect further evidence to support our findings, we plan to deploy this system, use it in educational settings, and evaluate its effectiveness by monitoring learners' progress in relation to each specific aspect of proficiency. Future work will also explore the use of multi-modal systems, such as the one presented in Tang et al. (2023), for assessing L2 speech in a similar fashion.

Limitations

The main limitation of this study is clearly the lack of ground truth analytic scores. The reader should keep in mind, however, that, as mentioned in Section 2, human analytic scoring is often an extremely difficult process, which might not produce completely reliable information. As evidence of this, the absence of publicly available L2 English learner datasets annotated with analytic scores speaks loud and clear and is not only an issue for the objectives of this paper, but for the whole scientific community.

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